

11

/ 13 LECTURES

NAT, DHCP and Local Network Communication

ARP's broadcast question, NAT/PAT's translation table, DHCP's DORA dance, and everything your humble home router secretly does.

ARP & Cache States

Gratuitous ARP

NAT · PAT State Table

Port Forwarding (DNAT)

DHCP DORA

Lease Renewal T1/T2

Home Router Anatomy

Full Local→Internet Journey

WHAT THIS LECTURE GIVES YOU

The machinery of everyday networks: how ARP turns IPs into MACs (with cache states and gratuitous ARP), how the NAT table lets five devices share one public IP (with a real state-table walkthrough), the DORA sequence that gives your laptop its address, lease renewal timers, the five devices hiding inside your home router, and a complete 10-step packet journey to 8.8.8.8.

STUDY-READY EDITION

CONCEPTS · DIAGRAMS · TABLES · EXAM PREP

LEARNING OBJECTIVES — AFTER THIS LECTURE YOU CAN

- Trace a full **ARP resolution** (cache → broadcast → reply → cache)
- Explain **gratuitous ARP** and proxy ARP
- Walk **NAT/PAT** flows through a state table; configure **port forwarding**
- Trace **DHCP DORA**, ports 67/68, and the T1/T2/T3 lease timeline
- Describe what a **home router** really is, and trace a full local→Internet round trip

Local Networks

ARP	NAT / PAT	DHCP	Home router
IP → MAC on the LAN	(priv IP:port) ↔ (pub IP:port)	DORA over UDP 67/68	router + switch + AP
broadcast Q, unicast A	accidental firewall	IP + mask + GW + DNS	+ DHCP + DNS + firewall
cache states REACHABLE→STALE	port forward = DNAT	renew at 50% · 87.5%	

1 ARP — Address Resolution Protocol

The gap: IP routes globally, Ethernet delivers locally by **MAC**. ARP bridges them: *"which MAC owns this IP on my LAN?"*

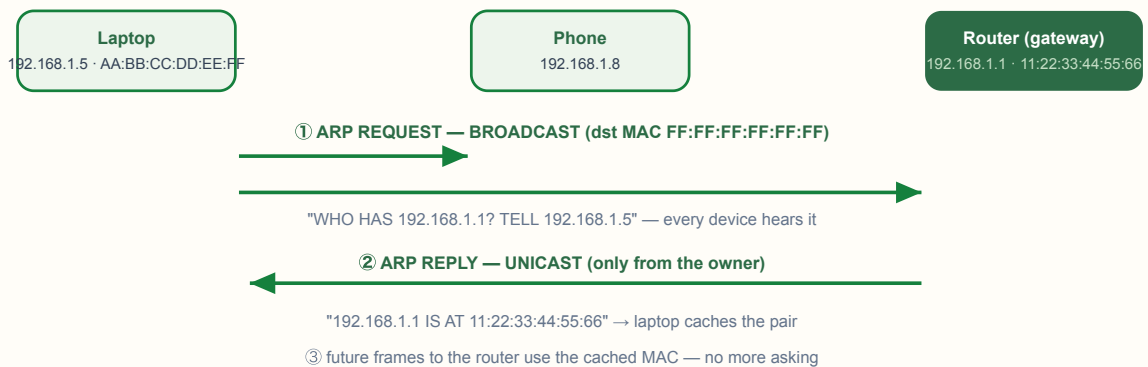


Figure 1. One broadcast question, one unicast answer, one cached mapping.

ARP COMMANDS

TERMINAL

```
arp -a          # view cache (all OSes)
ip neigh show   # Linux preferred — shows STATE too
ip neigh flush all # clear cache (root)
```

State	Meaning	State	Meaning
REACHABLE	confirmed recently (~30 s)	PROBE	sending unicast re-checks
STALE	unused; verify before next use	FAILED	no answer → entry deleted
DELAY	turned stale; probe pending	—	

Gratuitous ARP

A device ARPs for **its own IP**: (1) **conflict detection** — a reply means someone stole my IP!
(2) **cache refresh** — announce a new/failovered MAC to everyone.

Proxy ARP

A router answers ARP **on behalf of** devices on other subnets — legacy trick to make distant hosts look local.

2 NAT / PAT — the Translation Table

Setup: LAN 192.168.1.0/24, router WAN IP 203.0.113.5. The router keeps a live table mapping every conversation:

Private IP:Port	Public IP:Port	Talking to	Proto
192.168.1.5:52341	203.0.113.5:10001	142.250.195.46:80	TCP
192.168.1.8:34100	203.0.113.5:10002	142.250.195.46:80	TCP
192.168.1.12:45200	203.0.113.5:10003	54.230.45.100:443	TCP

Outbound — SNAT

```
laptop sends: 192.168.1.5:52341 →
142.250.195.46:80
router rewrites src: 203.0.113.5:10001
(+ creates entry)
```

Inbound reply — reverse lookup

```
google replies: → 203.0.113.5:10001
table: 10001 → 192.168.1.5:52341 →
delivers to laptop ✓
```

KEY

The port is the discriminator. Three devices share one public IP because each owns a distinct translated port. No table entry = **drop** — that's why NAT is an "accidental firewall" against unsolicited inbound traffic.

Port forwarding — DNAT

Static rule: public 27015 → 192.168.1.10:27015. Friends connect to your public IP:27015 and land on the game server.
Uses: game/web servers, cameras, SSH-at-home.

NAT traversal (P2P)

STUN: a server tells you your public IP:port → try direct. **TURN**: relay everything when STUN fails. **ICE**: tries STUN then TURN — how WebRTC video calls punch through NAT.

Broken by NAT	Why	Workaround
P2P apps	peers can't initiate inbound	STUN/TURN/ICE
VoIP (SIP)	embeds IPs inside payload	Application-Layer Gateway
FTP active mode	embeds IP+port in payload	passive mode / ALG

3 DHCP — the DORA Dance

DHCP hands a joining device everything it needs: **IP, subnet mask, default gateway, DNS server, lease time**. It runs over **UDP 67 (server) / 68 (client)** — UDP because the client has no IP yet and needs broadcast (TCP can do neither).

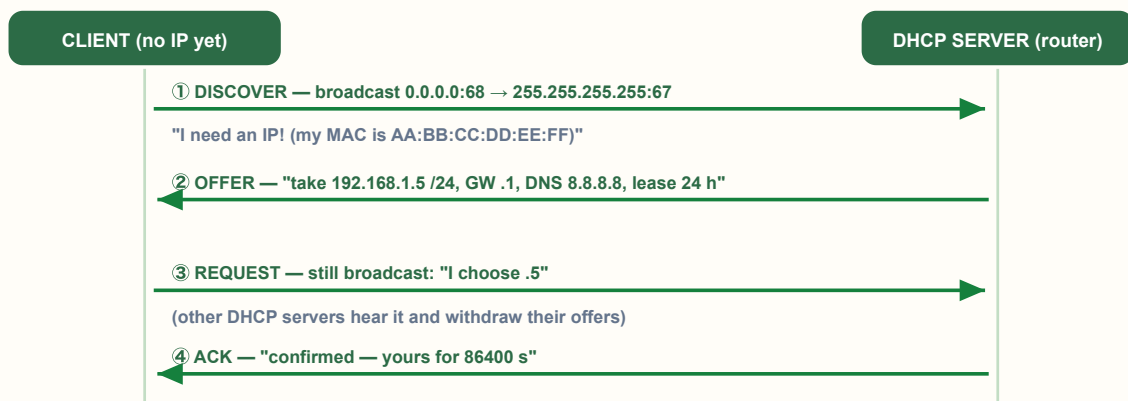


Figure 2. DORA — Discover, Offer, Request, Acknowledge. REQUEST stays broadcast on purpose (declining other servers).

Lease renewal timeline

Time	Event
T1 = 50% of lease	client unicasts REQUEST to the original server ("renew me")
T2 = 87.5%	no answer? broadcast REQUEST — any DHCP server may extend
T3 = 100%	lease expires — IP lost; full DORA restarts (brief outage)

24-hour lease example

T1 at 12 h (unicast renew) → T2 at 21 h (broadcast) → T3 at 24 h (expire + re-DORA). Large networks with many subnets use **DHCP relay agents** on routers to forward broadcasts to a central server.

4 Your Home Router Is Five Devices

Router (L3)

LAN ↔ WAN

Routes between LAN and Internet; performs the NAT we just studied.

DHCP Server

UDP 67

Runs DORA for every device that joins the LAN.

DNS Forwarder

UDP 53

Relays your queries to the ISP's or Google's resolvers (and caches).

L2 Switch

WIRED PORTS

MAC-based forwarding between wired LAN devices.

Wi-Fi AP

802.11

The wireless access point bridging phones/laptops into the LAN.

Firewall

+ NAT

Blocks unsolicited inbound (partly just NAT's table doing its thing).

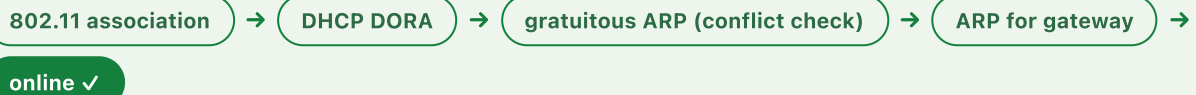
1 Router boots

WAN port runs its own DHCP DISCOVER to the ISP → gets public IP, gateway, DNS.

2 Router's DHCP starts

Serves private IPs from its pool to LAN clients.

New laptop joins the Wi-Fi — exact sequence



5 Complete Journey — Laptop (192.168.1.5) → 8.8.8.8

#	Step	Detail
1	Routing decision	8.8.8.8 $\notin$ 192.168.1.0/24 → send to default gateway 192.168.1.1
2	ARP gateway	(cache miss → broadcast) → router's MAC 11:22:33:44:55:66 cached
3	Frame to router	Eth: laptop→router · IP: 192.168.1.5 → 8.8.8.8
4	Router receives	strips Ethernet · LPM → default route → WAN interface
5	NAT (SNAT)	src 192.168.1.5:52341 → 203.0.113.5:10001 · table entry created
6	New frame out WAN	Eth: router-WAN→ISP-router · IP: 203.0.113.5 → 8.8.8.8
7	Across the Internet	router after router forwards toward 8.8.8.8 (Lecture 7)
8	Reply returns	8.8.8.8 → 203.0.113.5:10001
9	NAT reverse	10001 → 192.168.1.5:52341 · ARP laptop (cached) · deliver on LAN
10	Laptop demuxes	port 52341 → the browser's socket → app gets its data ✓

FULL CIRCLE This single journey uses: **subnetting** (L3/4) for the local-net check, **ARP**, **DHCP** (how we got the IP), **NAT**, **routing/LPM** (L7), and **ports** (L10) — everything except DNS/HTTP, which Lecture 13 adds.

LECTURE 11 — RAPID REVISION CARD

ARP: broadcast "who has IP X?" → unicast "X is at MAC" → cache. States:
REACHABLE→STALE→DELAY→PROBE→FAILED.

Gratuitous ARP: self-ARP = conflict check + cache refresh. Proxy ARP = router answers for others.

NAT table: (priv IP:port ↔ pub IP:port); port = discriminator; no entry → drop (accidental firewall).

Port forwarding = DNAT: fixed public port → private host:port. Traversal: STUN → TURN, chosen by ICE.

DORA: Discover(bcast) → Offer → Request(bcast!) → Ack. UDP 67 server / 68 client.

Lease: T1 50% unicast renew · T2 87.5% broadcast · T3 100% expire → re-DORA.

Home router = router + DHCP + DNS fwd + switch + AP (+firewall).

Boot/join order: router WAN-DHCP → serve LAN; laptop: Wi-Fi → DORA → gARP → ARP GW → online.

✦ Important Questions — Exam Practice

A. Multiple Choice

Q1. The ARP request is sent to MAC:

MCQ

- a) 00:00:00:00:00:00
- b) FF:FF:FF:FF:FF:FF
- c) the router's MAC
- d) 127.0.0.1

Q2. DHCP DISCOVER is sent from:

MCQ

- a) 192.168.1.5:68
- b) 0.0.0.0:68 to 255.255.255.255:67
- c) 127.0.0.1:67
- d) the client's old IP

Q3. DHCP REQUEST stays broadcast because:

MCQ

- a) TCP requires it
- b) other DHCP servers must hear that their offers are declined
- c) the client has no MAC
- d) routers demand it

Q4. A reply reaches 203.0.113.5:10001. The router forwards it using:

MCQ

- a) longest prefix match
- b) the NAT table (port 10001)
- c) ARP
- d) DNS

Q5. DHCP uses UDP because:

MCQ

- a) it's faster
- b) the client has no IP yet and needs broadcast — TCP offers neither
- c) UDP is encrypted
- d) TCP can't do ports

B. Short Answer (2–4 marks)

Q6. Why is the ARP request broadcast but the reply unicast?

SHORT

Q7. Two devices on one LAN get the same IP. What breaks and which ARP type reveals it?

SHORT

Q8. What uniquely identifies each connection in the NAT state table?

SHORT

Q9. Why can't an outsider connect to 192.168.1.5:8080 without port forwarding?

SHORT

Q10. Lease = 12 h. When are T1/T2/T3, and what happens at each?

SHORT

Q11. List five functions hidden inside a home "router".

SHORT

C. Long Answer (5–8 marks)

Q12. Trace a full ARP resolution with a diagram (cache check → broadcast → reply → cache), then explain gratuitous ARP's two purposes.

LONG

Q13. Draw the DHCP DORA sequence with addresses and ports at every message, explain why each message is broadcast or unicast, and the T1/T2/T3 renewal ladder for a 24 h lease.

LONG

Q14. With a NAT state table for two devices talking to the same server, explain outbound SNAT, inbound reverse translation, and why replies never get mixed up. Add a port-forwarding (DNAT) example.

LONG

Q15. Trace the complete local→Internet journey (laptop → 8.8.8.8 and back) through gateway ARP, LPM, SNAT, and demux — naming every header change.

LONG

ANSWER KEY

A1. b — broadcast reaches every LAN device.

A2. b — no IP yet, needs broadcast.

A3. b — implicit decline of other offers.

A4. b — NAT table lookup by translated port.

A5. b — no IP + broadcast need.

A6. Request must reach an unknown owner (everyone); the reply targets exactly one requester whose MAC is known from the question.

A7. Duplicate-IP chaos (intermittent reachability); gratuitous ARP by either device draws a reply → conflict detected.

A8. The translated (public) port — one entry per conversation.

A9. 192.168.1.5 isn't routable on the Internet, and even on the WAN IP there's no NAT entry for unsolicited inbound → dropped.

A10. T1 = 6 h unicast renew · T2 = 10.5 h broadcast · T3 = 12 h expire → re-DORA.

A11. Router + DHCP server + DNS forwarder + L2 switch + Wi-Fi AP (+ firewall).

A12–A15. Reproduce Figures 1–2, the state table, and the §5 journey table — annotate every header change for full marks.